A Review on Elementary Probability, Poisson Process & Markov Chains

Indian Academy of Sciences
Bangalore

Samriddha Ganguly
BS-MS Physics, IISER Bhopal
20 May, 2024.

Contents

Introduction 2
Elementary Ideas of Probability Theory 2
- Probability Spaces & Foundations 2
- Random Variables, Densities & Distributions 3
- Expectation Values, Variance & Moments 5
- Continuous Random Variables 7
- Expectation of Continuous Random Variables 8
- Moment Generating Functions and Characteristic Functions . 8
- Limit Theorems 9
- Important Inequalities in Probability 10

Brief Introduction to Stochastic Processes 11
Poisson Random Variable & Poisson Process 11
- The Poisson Random Variable 11
- The Poisson Process 13

Markov Chains & Random Walk 16
- Building the Idea 17
- The Chapman-Kolmogorov Equations 17
- Random Walks & Ergodic Markov Chains 19

Conclusion 21
Appendix 22
- Appendix 1: Generating a Random Walk using Mathematica . 22
- Appendix 2: Proof of Weak Law of Large Numbers 23

References 23

Introduction

Probability theory is the branch of Mathematics dealing with Random or chance phenomena. The study of Probability is very important in Physics as often we try to characterize random processes. Fields like Quantum Mechanics & Statistical Physics base their foundations on Probability & Statistics. Apart from application in Physics, Probability theory finds various applications in Economics & Finance, Biology, Cryptography & Encryption just to name a few fields.

In this short review I plan to introduce and explain to the best of my knowledge two key mathematical structures in Probability Theory: Poisson Processes & Markov Chains. I will also delve into some applications of the same specially those related to Physics & related fields.

Elementary Ideas of Probability Theory

Before we begin let us briefly look at some of the pre-requisite knowledge required to proceed further.

Probability Spaces & Foundations

We can take different approaches as to defining the idea of probability of any event.
Let us try to define the probability of an event in terms of its relative frequency. Let us consider an experiment with a sample space S and consider $\exists$ an event $E \in S$. Let $n(E)$ denote the number of times event E occurs in first n iterations of the experiment. Then we can mathematically define the Probability of event E occurring, $P(E)$ is :

$$P(E) = \lim_{n\to\infty} \frac{n(E)}{n}$$

That is, $P(E)$ is defined as the (limiting) proportion of time that E occurs. It is thus the limiting frequency of E.
The problem arises is we do not know that this limit always converges or what if it converges but is that value same for all new iterations of the experiment? That is, how do we know that $\frac{n(E)}{n}$ will converge to some constant

limiting value that will be the same for each possible sequence of repetitions of the experiment? This gives rise to need for certain assumptions and a more "set-theoretic" approach is used.

For the new definition let us first define what is a σ-algebra. $\nexists$ Consider a set Ω. Now a **non-empty collection of subsets of** Ω, let $\mathcal{A}$ is called a σ-field (or σ-algebra) if the following conditions hold:

- If $A \in \mathcal{A}$ then $A^c \in \mathcal{A}$
- If $A_n \in \mathcal{A}$, n=1,2,... then $\bigcup_{n=1}^{\infty} A_n$ and $\bigcap_{n=1}^{\infty} A_n$ are both in $\mathcal{A}$

From this we can thus define the Probability Measure P on a σ-field of subsets $\mathcal{A}$ of a set Ω as a function,

$$P : \mathcal{A} \to \mathbb{R}$$

And, obeys the following Axioms which make it a useful function:

- $P(\Omega) = 1$
- $P(A) \geq 0 \ \forall \ A \in \mathcal{A}$
- If A_n, n=1,2,3,... are mutually disjoint sets in $\mathcal{A}$, then we may say,

$$P(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} P(A_n)$$

A probability space denoted by $(\Omega, \mathcal{A}, P)$ is a set Ω, a σ-field of subsets $\mathcal{A}$ on Ω, and a probability measure P defined on $\mathcal{A}$.

Other propositions relating to the Probabilities of events can be derived using logic from Set Theory. From here we will look to build more on the concepts of Probability and introduce more mathematical structures it add rigour to our study of random phenomena

Random Variables, Densities & Distributions

Definition: A discrete real-valued random variable X on a probability space $(\Omega, \mathcal{A}, P)$ is a function,

$$X : \Omega \to \mathbb{R}$$

where we see, the range is a finite or countably infinite subset of $\mathbb{R}$, such that $\{\omega : X(\omega) = x_i\}$ is an event $\forall\ i\ \&\ \omega \in \Omega$.

Having defined what is a Random Variable we can explore another important tool called the probability density function or the probability mass function of a random variable X.

Definition: The function $f : \mathbb{R} \to \mathbb{R}$ defined by,

$$f(x) := P(X = x)$$

is called the discrete density function of X. A number x is called a possible value of X if $f(x) > 0$.
After having a proper definition we can list down a few important properties of the discrete density function f:

- $f(x) \geq 0\ \forall\ x \in \mathbb{R}$
- $\{x : f(x) \neq 0\}$ is a finite or countably infinite subset of $\mathbb{R}$. Let $\{x_1, x_2, ...\}$ denote this set. Then,
- $\Sigma_i f(x_i) = 1$

We can now alternately define a real valued function f to be a discrete density function if it satisfies all these three properties.
Now we can try to define another function which is actually more useful later.

Definition: The distribution function $F(t)$ of a discrete random variable X is defined by

$$F(t) = P(X \leq t) = \sum_{x \leq t} f(x)$$

It is actually non-decreasing step function. We can now generalize this notion of random variables to r-dimensional spaces.

Generalization: A discrete r-dimensional random vector X on a probability space $(\Omega, \mathcal{A}, P)$ is a function,

$$X : \Omega \to \mathbb{R}^r$$

where we see, the range is a finite or countably infinite number of values $x_1, x_2, ...$ such that, $\{\omega : X(\omega) = x_i\}$ is an event $\forall\ i\ \&\ \omega \in \Omega$.

Similarly, we can define the r-dimensional discrete density function f of X as follows:

$$f(x_1, ..., x_r) = P(X_1 = x_1, ..., X_r = x_r)$$

or more simply,

$$f(x) := P(X = x), x \in \mathbb{R}^r$$

Briefly I will state that this is called **joint density function** of all these random variables and each density function of each random variable X_i is called its **marginal density function**.

Condition for Independence: Let us consider two random variables X & Y, and take their marginal densities, f_X & f_Y respectively. If they are independent then we can write their joint density function f as:

$$f(x, y) = f_X(x) f_Y(y)$$

Finally we conclude this segment with a very useful tool for the future known as the Probability Generating Function.

Definition: Let X be a non-negative integer-valued random variable. The probability generating function Φ_X of X is defined as:

$$\Phi_X(t) = \sum_{x=0}^{\infty} f_X(x) t^x, \ -1 \leq t \leq 1$$

The concepts in this part will be largely used as we look into the cases for continuous random variables and this section will be frequently referred.

Expectation Values, Variance & Moments

Definition: Let X be a discrete random variable having density f. If $\sum_j |x_j| f(x_j) < \infty$, then we say X has a finite expectation and it is defined as:

$$E(X) := \sum_{j=1}^{\infty} x_j f(x_j)$$

It is sometimes also called the mean of X and denoted by μ.

Properties of Expectation: We will now try to understand more about this expectation value through its properties.

- Let X be a discrete random vector having density f and let φ be a real valued function defined on $\mathbb{R}^r$. Then the random variable $Z = \varphi(X)$ has finite expectation iff $\sum_x |\varphi(x)| f(x) < \infty$ and it is defined as:

$$E(Z) := \sum_x \varphi(x) f(x)$$

- Let X and Y be two random variables having finite expectation then,

  (i) If c is a constant and $P(X = c) = 1$, then $E(x) = c$.

  (ii) If c is a constant, then cX has finite expectation and $E(cX) = cE(X)$.

  (iii) $X + Y$ has finite expectation and,

$$E(X + Y) = E(X) + E(Y)$$

  (iv) If $P(X \geq Y) = 1$, then $E(X) > E(Y)$. Also, $E(X) = E(Y)$ iff $P(X = Y) = 1$

  (v) $|E(X)| \leq E|X|$

  Proof of these have been omitted as we are more focused on their applications.

- Let X be a random variable such that for some constant M, $P(|X| \leq M) = 1$. Then X has a finite expectation and $|E(X)| \leq M$.

- Let X and Y be two independent random variables having finite expectation then XY has finite expectation and,

$$E(XY) = E(X)E(Y)$$

Moments: Let X be a discrete random variable and let $r \geq 0$ be an integer. We say that X has a moment of order r if X^r has a finite expectation. It is then defined as the rth moment as $E(X^r)$.

Similarly we can say if X has a moment of order r then the rth moment of $X - \mu$ is called the rth central moment of X. Therefore,

$$E(X^r) = \sum_x x^r f(x)$$

$$E((X-\mu)^r) = \sum_x (x-\mu)^r f(x)$$

Definition: Let X be a random variable having a finite second moment then we can define the variance of X as:

$$Var(X) := E((X - E(X))^2)$$

So we can say Variance of X is basically the 2nd central moment of X.
We often denote variance by σ^2, and we define the non-negative number $\sigma = \sqrt{Var(X)}$ as the standard deviation of X.

Continuous Random Variables

Firstly let us define two important structures.

Definition 1: A random variable X on a probability space $(\Omega, \mathcal{A}, P)$ is a real-valued function $X(\omega),\ \omega \in \Omega$, such that for $-\infty < x < \infty,\ \{\omega : X(\omega) = x\}$ is an event.

Definition 2: The distribution function F of a random variable X is the function:

$$F(x) = P(X \leq x), \quad -\infty < x < \infty$$

Now we can start to build the logic towards the continuous random variable by defining it.

Definition 3: A random variable X is called a continuous random variable if

$$P(X = x) = 0, \quad -\infty < x < \infty$$

i.e. we can say a random variable X is continuous if the distribution function F describing X is a continuous function.

We can now also define the density function(w.r.t integration) as basically a non-negative function f such that,

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Now note if f is the density function then the distribution function F is defined as:

$$F(x) = \int_{-\infty}^{x} f(y)dy, \quad -\infty < x < \infty$$

.

Thus we notice for computation purposes, the easiest way to obtain the density function is:

$$f = F'$$

Now let us also define conditional densities of continuous random variables. We will state the important formulae and leave it at that citing more important matter to be covered. Let X & Y be continuous random variables having joint density function f. The conditional density $f_{Y|X}$ is defined by:

$$f_{Y|X}(y|x) = \begin{cases} \frac{f(x,y)}{f_X(x)}, & 0 < f_X(x) < \infty \\ 0, & elsewhere \end{cases}$$

We use this to modify the Bayes' Rule for Continuous cases:

$$f_{X|Y}(x|y) = \frac{f_X(x)f_{Y|X}(y|x)}{\int_{-\infty}^{\infty} f_X(x)f_{Y|X}(y|x)dx}$$

Expectation of Continuous Random Variables

Definition: Let X be a discrete random variable having density f. If $\int_{-\infty}^{\infty} |x|f(x)dx < \infty$, then we say X has a finite expectation and it is defined as:

$$E(X) := \int_{-\infty}^{\infty} xf(x)dx$$

Similarly, we can look to generate moments of continuous random variables by replacing the summation notation with the integral as it reflects the continuous nature of the distribution.

Moment Generating Functions and Characteristic Functions

Definition: The moment generating function $M_X(t)$ of a random variable X is defined by:

$$M_X(t) = E(e^{tX})$$

From which we can generate moments of all order of X using the following relation:

$$EX^n = \frac{d^n}{dx^n} M_X(t)|_{t=0}$$

Definition: The characteristic function of a random variable X is defined as:

$$\varphi_X(t) = E(e^{itX}), \quad -\infty < t < \infty$$

By definition of the characteristic function, which is basically just the Fourier Transform of the density function, we can thus, regenerate the density function by an Inverse Fourier Transform.

$$f_X(k) = \frac{1}{2\pi} = \int_{-\pi}^{\pi} e^{-ikt} \varphi_X(t) dt$$

This is known as the "Inversion Formula".

Continuity Theorem: Let X_n, $n \geq 1$, and X be random variables such that:

$$\lim_{n\to\infty} \varphi_{X_n}(t) = \varphi_X(t), \quad -\infty < t < \infty$$

. Then,

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

at all points x where F_X is continuous.

Limit Theorems

1) The Weak Law of Large Numbers: Let $X_1, X_2, ...$ be independent, identically distributed (i.i.d) random variables having finite mean μ & set $S_n = X_1 + ... + X_n$. Then for any $\varepsilon > 0$,

$$\lim_{n\to\infty} P\Big(\Big|\frac{S_n}{n} - \mu\Big| > \varepsilon\Big) = 0$$

Proof of this theorem is included in the appendix.

2) The Central Limit Theorem: Let $X_1, X_2, ...$ be i.i.d random variables each having mean μ and finite non-zero variance σ^2, then:

$$\lim_{n\to\infty} P\Big(\Big|\frac{S_n - n\mu}{\sigma\sqrt{n}} - \mu\Big| \leq x\Big) = \Phi(x), \quad -\infty < x < \infty$$

where $\Phi(x)$ denotes the standard normal distribution function.

3) The Strong Law of Large Numbers: If $X_1, X_2, ...$ are i.i.d random variables with mean μ, then:

$$P\{\lim_{n\to\infty} \frac{(X_1 + ... + X_n)}{n} = \mu\} = 1$$

See Appendix

Important Inequalities in Probability

1) Markov Inequality: If X be a non-negative random variable, then for any $a > 0$

$$P(X \geq a) \leq \frac{E(X)}{a}$$

2) Chebyshev's Inequality: Let X be a random variable with mean μ and finite variance σ^2. Then for any real number $a > 0$

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Brief Introduction to Stochastic Processes

A Stochastic Process may be thought of as a collection of random variables. It may be vague to think about it like this, however it is also required to keep in mind that when we think of such processes we usually refer to processes with enough additional structure so as to give rise to interesting and useful results. Poisson Processes, Markov Chains and consequently random walks are all types of stochastic processes. We will now try to give "mathematical" definition for a stochastic process.

Definition: Let $\mathcal{T}$ be a subset of $[0, \infty)$. A collection of random variables $\{X_t\}_{t \in \mathcal{T}}$, i.e. each random variable is indexed by a member of $\mathcal{T}$, is called a stochastic process. When $\mathcal{T} = \mathbb{N}$ (or $\mathcal{T} = \mathbb{N}_{\nvdash}$), that is, if $\mathcal{T}$ is a countable set $\{X_t\}_{t \in \mathcal{T}}$ is said to be a discrete-time process, and when $\mathcal{T} = [0, \infty)$, that is, $\mathcal{T}$ is not a countable set, it is called a continuous-time process.

In the Appendix, I have included the process to construct a simple random walk which is one of the elementary stochastic processes.

Poisson Random Variable & Poisson Process

In this section we will delve into our first focus which is the Poisson Process. We will aim to provide necessary background, definition and formulation of a Poisson process and process of constructing one in a mathematical way.

The Poisson Random Variable

A random variable X that takes on one of the values $0, 1, 2, ...$ is said to be a Poisson random variable with parameter λ if, for some $\lambda > 0$,

$$f(x) = P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!} \quad x = 0, 1, 2, ...$$

The Poisson probability distribution was introduced by Sim´ eon Denis Poisson in a book he wrote regarding the application of probability theory to lawsuits, criminal trials, and the like. This book, published in 1837, was entitled *Recherches sur la prob abilit´e des jugements en mati'ere criminelle et en mati'ere civile* (Investigations into the Probability of Verdicts in Criminal

and Civil Matters).
The Poisson random variable has such a diverse range of uses due to its unique property of being a binomial approximation. Below we will see the proof of this.

We will approach this proof a little differently. Important fact we remember from our knowledge of discrete random variables is that any Bernoulli Random Variable is basically a Binomial distribution with parameters $(1, p)$.

Thus, now consider that we have n such Bernoulli Trials each being a random variable X_i probability p of a success. Now let us consider the fact that if we sum all these to generate a new random variable $S_n = X_1 + X_2 + ...X_n$, where this is nothing but a Binomial Distribution with parameter (n, p). Now, probability of having $S_n = k$ successes in n trials is,

$$P(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Now, let us consider a parameter $\lambda > 0$, $\lambda = np$ and thus, $p = \frac{\lambda}{n}$. We will use this in the above equation.

$$P(S_n = k) = \frac{\lambda^k}{k!} \frac{n!}{(n-k)!n^k} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}$$

Now for the approximation we consider $n \to \infty$. So,

$$\lim_{n\to\infty} P(S_n = k) = \frac{\lambda^k}{k!} \lim_{n\to\infty} \frac{n!}{(n-k)!n^k} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}$$

We will try to solve this limit by dividing it into some other limits. Firstly,

$$\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e$$

which implies that if we take $x = -\frac{n}{\lambda}$,

$$\lim_{n\to\infty} \left(1 + \frac{\lambda}{n}\right)^n = e^{-\lambda}$$

Also for the next limit simple consideration takes us to the answer,

$$\lim_{n\to\infty} \left(1 + \frac{\lambda}{n}\right)^{-k} = 1$$

Now what we are left to solve is,

$$\lim_{n\to\infty} \frac{n!}{(n-k)!n^k}$$

By expanding,

$$\lim_{n\to\infty} \frac{n(n-1)(n-2)...(n-k+1)}{n^k} = \lim_{n\to\infty} \left(\frac{n}{n}\right)\left(\frac{n-1}{n}\right)\left(\frac{n-2}{n}\right)...\left(\frac{n-k+1}{n}\right)$$

We hence observe that all these k terms approach 1 as $n \to \infty$, thus this limit approaches 1 also. Finally we can replace our limits in the original distribution.

$$\lim_{n\to\infty} P(S_n = k) = \frac{\lambda^k}{k!}e^{-\lambda}$$

Thus we conclude, if n independent trials, each of which results in a success with probability p, are performed, then, when n is large and p is small enough to make np moderate, the number of successes occurring is approximately a Poisson random variable with parameter $\lambda = np$. This value λ (which is equal the expected number of successes) will usually be determined empirically.

The Poisson Process

Before we define the Poisson Process we need to first state a definition. A function f is said to be $o(h)$ if:

$$\lim_{h\to 0} \frac{f(h)}{h} = 0$$

That is, f is $o(h)$ if, for small values of h, $f(h)$ is small even in relation to h. Suppose now that "events" are occurring at random points at time, and let $N(t)$ denote the number of events that occur in the time interval $[0, t]$. The collection of random variables $N(t), t \geq 0$ is said to be a Poisson process having rate λ, $\lambda > 0$, if:

- $N(0) = 0$.
- The numbers of events that occur in disjoint time intervals are independent.(Independent Increment assumption)

- The distribution of the number of events that occur in a given interval depends only on the length of that interval and not on its location. (stationary increment assumption)
- $P(N(h) = 1) = \lambda h + o(h)$
- $P(N(h) \geq 2) = o(h)$

The stationary increment assumption, states that the probability distribution of $N(t+s) - N(t)$ is the same for all values of t.

Lemma 1:

For a Poisson process with rate λ,

$$P(N(t) = 0) = e^{-\lambda t}$$

Proof: Let $P_0(t) = P(N(t) = 0)$. We derive a differential equation for $P_0(t)$ in the following manner:

$$\begin{aligned} P_0(t+h) &= P(N(t+h) = 0) \\ &= P(N(t) = 0, N(t+h) - N(t) = 0) \\ &= P(N(t) = 0)P(N(t+h) - N(t) = 0) \\ &= P_0(t)[1 - \lambda h + o(h)] \end{aligned}$$

Here, we have only used considerations from the definition of a Poisson Process: particularly last two points to arrive at this expression. Now, we can try to find the derivative of $P_0(t)$:

$$\lim_{h \to 0} \frac{P_0(t+h) - P_0(t)}{h} = \lim_{h \to \infty} = -\lambda P_0(t) + \frac{o(h)}{h}$$

Thus we see,

$$P_0'(t) = -\lambda P_0(t)$$

By integrating we get,

$$log(P_0(t)) = -\lambda t + c$$

But we know, $P_0(0) = P(N(0) = 0) = 1$, so we say, $c = 0$. Hence we can state finally,

$$P_0(t) = e^{-\lambda t}$$

$\square$

For a Poisson process, let T_1 denote the time the first event occurs. Further, for $n > 1$, let T_n denote the time elapsed between the $(n-1)st$ and the nth event. The sequence $\{T_n, n = 1, 2, ...\}$ is called the sequence of interarrival times. For instance, if $T_1 = 5$ and $T_2 = 10$,then the first event of the Poisson process would have occurred at time 5 and the second at time 15.
We shall now determine the distribution of the T_n. To do so, we first note that the event $T_1 > t$ takes place iff no events of the Poisson process occur in the interval $[0, t]$, thus,

$$P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}$$

From this we can easily make-out the fact that T_1 has an exponential distribution with mean $\frac{1}{\lambda}$.Now,

$$P(T_2 > t) = E[P(T_2 > t|T_1)]$$

However,

$$\begin{aligned} P(T_2 > t|T_1 = s) &= P(0\, events\, in\, (s, s+t]|T_1 = s) \\ &= P(0\, events\, in\, (s, s+t]) \\ &= e^{-\lambda t} \end{aligned}$$

where the last two equations followed from the assumptions about independent and stationary increments. From the preceding, we conclude that T_2 is also an exponential random variable with mean $\frac{1}{\lambda}$ and, furthermore, that T_2 is independent of T_1. Repeating the same argument yields our next proposition,

Proposition 1: $T_1, T_2, ...$ are independent exponential random variables, each with mean $\frac{1}{\lambda}$.

Another quantity of interest is S_n, the arrival time of the nth event, also called the waiting time until the nth event. It is easily seen that

$$S_n = \sum_{i=1}^{n} T_i \quad n \geq 1$$

hence, from Proposition 1 and the results of exponential variables, it follows that S_n has a Γ distribution with parameters n and λ. That is, the probability

density of S_n is given by:

$$f_{S_n}(x) = \frac{\lambda e^{-\lambda x}(\lambda x)^{n-1}}{(n-1)!} \quad x \geq 0$$

Now we can finally proceed to prove that $N(t)$ is a Poisson random variable with mean λt.

Theorem 1:

For a Poisson process with rate λ,

$$P(N(t) = n) = \frac{e^{-\lambda t}(\lambda t)^n}{n!}$$

Proof: Note that the nth event of the Poisson process will occur before or at time t if and only if the number of events that occur by t is at least n. That is,

$$N(t) \geq n \Longleftrightarrow S_n \leq t$$

So,

$$\begin{aligned} P(N(t) = n) &= P(N(t) \geq n) - P(N(t) \geq n+1) \\ &= P(S_n \leq t) - P(S_{n+1} \leq t) \\ &= \int_0^t \frac{\lambda e^{-\lambda x}(\lambda x)^{n-1}}{(n-1)!} dx - \int_0^t \frac{\lambda e^{-\lambda x}(\lambda x)^n}{(n)!} dx \end{aligned}$$

But the integration-by-parts formula $\int u dv = uv - \int v du$ with $u = e^{-\lambda x}$ and $dv = \lambda[(\lambda x)^{n-1}/(n-1)!]dx$ yields,

$$\int_0^t \frac{\lambda e^{-\lambda x}(\lambda x)^{n-1}}{(n-1)!} dx = e^{-\lambda t}\frac{(\lambda t)^n}{n!} + \int_0^t \frac{\lambda e^{-\lambda x}(\lambda x)^n}{(n)!} dx$$

Hence we can conclude this proof $\square$

We conclude this section stating an interesting result: The Poisson process $N(t)$ has the property:

$$P(N(t_n) = j_n | N(t_{n-1}) = j_{n-1}, ...N(t_1) = j_1) = P(N(t_n) = j_n | N(t_{n-1}) = j_{n-1})$$

This basically implies that the conditional probability of any $N(t_i)$ given all previous random variables, is only dependent on the immediately preceding random variable. This interesting property is called the **Markov Property** and we will be learning about this in the next section.

Markov Chains & Random Walks

Building the Idea

In many real life situations we notice that often phenomena we observe are in fact not truly independent. Let us take the example of weather forecasting. We cannot say that weather of one day is totally independent of the previous days. However, we can say that the conditions of the previous day may have some bearing on the next day. This is essentially the idea behind Markov Chains. Now, we will try to define this mathematically.

Definition: A stochastic process $\{X_n\}_{n\in\mathbb{N}_0}$ taking values in a countable state space S is called a Markov chain (or said to have the Markov property) if

$$P[X_{n+1} = i_{n+1}|X_n = i_n, X_{n-1} = i_{n-1}, ..., X_1 = i_1, X_0 = i_0] = P[X_{n+1} = i_{n+1}|X_n = i_n]$$

So basically we can interpret this as: the conditional distribution of any future state X_{n+1}, given the past states $X_0, X_1, ...X_{n-1}$ and present state X_n is independent of the past states and is solely dependent on the present state. Now let P_{ij} denote the probability that the process will when in state i, next make a transition to state j. Since probabilities are non-negative and since the process must make a transition into some state we have that:

$$P_{ij} \geq 0, \quad i, j \geq 0; \quad \sum_{j=0}^{\infty} P_{ij} = 1, \quad i = 0, 1, ...$$

Let P denote the matrix of one-step transition probabilities P_{ij}, so that

$$\begin{pmatrix} P_{00} & P_{01} & P_{02} & \dots \\ P_{10} & P_{11} & P_{12} & \dots \\ \vdots & & & \\ P_{i0} & P_{i1} & P_{i2} & \dots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

Such an array is called a The one-step transition matrix. Knowledge of the transition probability matrix and of the distribution of X_0 enables us, in theory, to compute all probabilities of interest.

The Chapman-Kolmogorov Equations

Having defined the one-step transition probability, let us move forward with our logic and try to define the two-step transition probability. We can also define the two-stage transition probability P_{ij}^2 that a system presently in state i will be in state j after two additional transitions. That is,

$$P_{ij}^2 = P(X_{m+2} = j | X_m = i)$$

The P_{ij}^2 can be computed from the P_{ij} as follows:

$$\begin{aligned}
P_{ij}^2 &= P(X_2 = j | X_0 = i) \\
&= \sum_{k=0}^{M} P(X_2 = j, X_1 = k | X_0 = i) \\
&= \sum_{k=0}^{M} P(X_2 = j | X_1 = k, X_0 = i) P(X_1 = k | X_0 = i) \\
&= \sum_{k=0}^{M} P_{kj} P_{ik}
\end{aligned}$$

In general, we define the $n-$stage transition probabilities, P_{ij}^n as:

$$P_{ij}^n = P(X_{m+n} = j | X_m = i)$$

Proposition 2: The Chapman–Kolmogorov equations

$$P_{ij}^n = \sum_{k=0}^{M} P_{ik}^{(r)} P_{kj}^{(n-r)}$$

Proof:

$$\begin{aligned}
P_{ij}^n &= P(X_n = j | X_0 = i) \\
&= \sum_{k=0}^{M} P(X_n = j, X_r = k | X_0 = i) \\
&= \sum_{k=0}^{M} P(X_n = j | X_r = k, X_0 = i) P(X_r = k | X_0 = i) \\
&= \sum_{k=0}^{M} P_{kj}^{(n-r)} P_{ik}^{(r)}
\end{aligned}$$

□

Random Walks & Ergodic Markov Chains

An example of a Markov chain having a countably infinite state space is the random walk, which tracks a particle as it moves along a one-dimensional axis. Suppose that at each point in time the particle will move either one step to the right or one step to the left with respective probabilities p and $1-p$. That is, suppose the particle's path follows a Markov chain with transition probabilities:

$$P_{i.i+1} = p = 1 - Pi, i-1 \quad i = 0, \pm 1, ...$$

If the particle is at state i, then the probability that it will be at state j after n transitions is the probability that $(n-i+j)/2$ of these steps are to the right and $n - [(n-i+j)/2] = (n+i-j)/2$ are to the left. Since each step will be to the right, independently of the other steps, with probability p, it follows that the above is just the binomial probability.

$$P_{ij}^n = \binom{n}{(n-i+j)/2} p^{(n-i+j)/2} (1-p)^{(n+i-j)/2}$$

We can also re-write this expression as:

$$P_{i,i+2k}^{2n} = \binom{2n}{n+k} p^{(n+k)} (1-p)^{(n-k)} \quad k = 0, \pm 1, \pm 2, ..., \pm n$$

$$P_{i,i+2k+1}^{2n+1} = \binom{2n+1}{n+k+1} p^{(n+k+1)} (1-p)^{(n-k)} \quad k = 0, \pm 1, \pm 2, ..., \pm n, -(n+1)$$

We can use this, although denoting the conditional probabilities we can use it to find the unconditional probabilities too as follows:

$$\begin{aligned} P(X_n = j) &= \sum_i P(X_n = j | X_0 = i) P(X_0 = i) \\ &= \sum_i P_{ij}^n P(X_0 = i) \end{aligned}$$

For a large number of Markov chains, it turns out that P_{ij}^n converges, as $n \to \infty$, to a value π_j that depends only on j. That is, for large values of n, the probability of being in state j after n transitions is approximately equal to π_j, no matter what the initial state was. It can be shown that a sufficient condition for a Markov chain to possess this property is that, for some $n > 0$,

$$P_{ij}^n > 0$$

$\forall\ i, j = 0, 1, ..., M$
Markov Chains that satisfy this equation are said to be ergodic. Since Proposition 2 yields

$$P_{ij}^{n+1} = \sum_{k=0}^{M} P_{ik}^n P_{kj}$$

it follows, by letting $n \to \infty$, that, for ergodic chains,

$$\pi_j = \sum_{k=0}^{M} \pi_k P_{kj}$$

Furthermore, since $\sum_{k=0}^{M} P_{ij}^n$, we also obtain, by letting $n \to \infty$:

$$\sum_{k=0}^{M} \pi_j = 1$$

We will conclude our discussions by stating a theorem without proof summarizing the previous section.

Theorem 2:

For an ergodic Markov chain,

$$\pi_j = \lim_{n \to \infty} P_{ij}^n$$

exists, and the π_j, $0 \leq j \leq M$, are the unique non-negative solutions of

$$\pi_j = \sum_{k=0}^{M} \pi_k P_{kj}$$

$$\sum_{k=0}^{M} \pi_j = 1$$

Conclusion

All random walks, Wiener process (Brownian Motion), Poisson process, etc. are all examples of Markov Chains. The remarkable implications of Markov Chains specifically in studying Stochastic Dynamic systems is particularly a useful tool for Physicists. I will want to delve further into the realm of Stochastic Mechanics, ,particularly focussing on the fundamentals building up to diffusion processes, derivation of Schrödinger's Equation from Diffusion processes i.e. the stochastic interpretation of quantum mechanics.

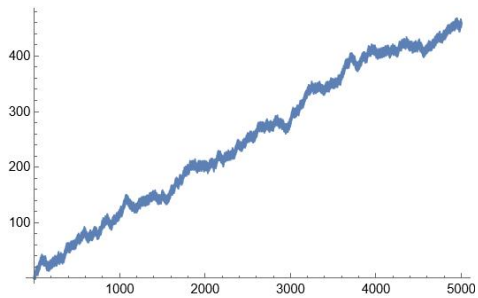

Figure 1: This graph illustrates p=0.55 one dimensional Random Walk

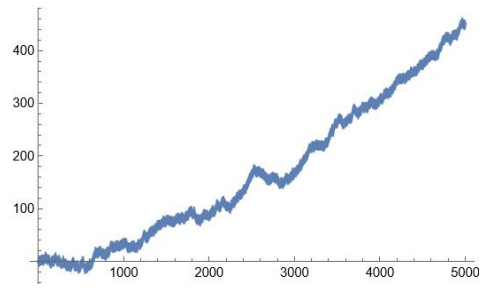

Figure 2: This is the second iteration of same p=0.55 Random Walk

Appendix

Appendix 1: Generating a Random Walk using Mathematica

This is something extra I gathered through an introductory course in Computational Physics during my 4th semester.Here, we will show how to generate a biased random walk and see how it approximates to a normal distribution using the software Mathematica.

```
a={0};
p=k;
Do[AppendTo[a,a[[n-1]]+1-2UnitStep[RandomReal[]-p]],{n,2,5000}]
ListPlot[a,Joined->True]
```

This code illustrates a biased random procces where k can be any biased probability value between $[0, 1]$. By using formula that variance is just $\sigma^2 = 4Npq$ & mean is $\mu = Npq$, where N is the number of iterations we can generate graphs as shown in Figure 1 & Figure 2. Now we will give the code for approximation using the Central Limit Theorem:

```
nmax=1000;
binsize=10;

histdata=Histogram[Table[Table[1-2UnitStep[RandomReal[]-0.55],{nmax}]//Total,{5000

Show[histdata, Plot[1/(2*Pi*nmax*4*0.55*0.45)^(1/2) Exp[-(x-2*nmax(0.55-0.5))^2/(2
```

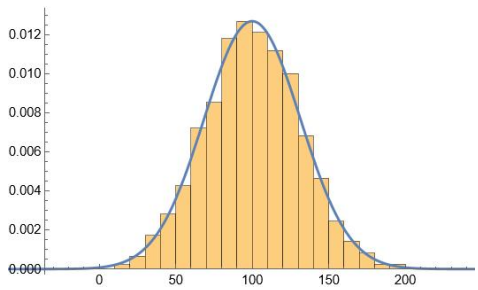

Figure 3: This graph illustrates the Random Walk PDF approximated with the Normal Distribution

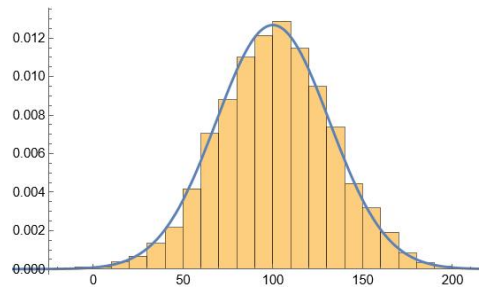

Figure 4: This is the second iteration of same Random Walk PDF

From this we get the following outputs for 2 iterations in Figure 3 & Figure 4 Thus we can conclude that the Random Walk Probability Distribution Function indeed approximates to a Normal Distribution.

Appendix 2: Proof of Weak Law of Large Numbers

We shall prove the theorem only under the additional assumption that the random variables have a finite variance σ^2. Now, since,

$$E\left[\frac{X_1 + X_2 + ... + X_n}{n}\right] = \mu \quad and \quad Var\left[\frac{X_1 + X_2 + ... + X_n}{n}\right] = \frac{\sigma^2}{n}$$

It then follows from Chebyshev's Inequality that:

$$P\left[\left|\frac{X_1 + X_2 + ... + X_n}{n} - \mu\right| \geq \varepsilon\right] \leq \frac{\sigma^2}{n\varepsilon^2}$$

and thus, the result is proven. □

Proofs of *Strong Law of Large Numbers* & the *Central Limit Theorem* can be found in great detail in chapter 8 of *"A First Course in Probability"* (8th ed) by Sheldon Ross, hence I find no need to elaborate here.

References

[1] Hoel, Port, Stone (1971) *Introduction to Probability Theory*, Houghton Mifflin Co.

[2] Sheldon Ross (2010) *A First Course in Probability (Ninth Edition)*, Pearson.

[3] Sheldon Ross (1995) *Stochastic Processes (Second Edition)*, John Wiley & Sons, Inc.

[4] Prof. Ulrich Schwarz (2019) *Stochastic Dynamics*, Heidelberg University, Department of Physics and Astronomy.

[5] Prof. Hao Wu (2015) *18.445 Introduction to Stochastic Processes*, MIT OpenCourseWare.

[6] Prof. Gordan Zitkovic (2019) *M362M Introduction to Stochastic Processes*, University of Texas at Austin.

[7] D.P. Bertsekas & John N. Tsitsiklis (2008) *Introduction to Probability (Second Edition)*, Athena Scientific.

[8] Prof. Prahlad Vaidyanathan (2024) *MTH202O Probability & Statistics*, IISER Bhopal.